Can Calendar Signals Help Predict Next-Day Stock Direction for Apple?

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Can Calendar Signals Help Predict Next-Day Stock Direction for Apple? – Context

- **Context:** This project investigates and analyzes whether calendar signals along with price and volume features can help predict next-day stock direction. The goal of this analysis is to gain a further understanding on if, how, and where such calendar effects exist that may help predict future stock performance.
- **Dataset:** The dataset covered the last 5 years of stock data involving Apple. The dataset was obtained from Yahoo Finance and contained the following columns originally: ‘Date’, ‘Open’, ‘High’, ‘Low’, ‘ADJ Close’, and ‘Volume’.
- **Machine Learning Models Used:** Three machine learning models were used in this project: Gradient Boosting, Random Forest, and Logistic Regression. A time-series cross validation framework was used to prevent bias in the future, as well as any data leakage. The models were evaluated by accuracy, ROC-AUC, and Log-Loss. Because accuracy can be misleading at times due to inefficient evidence of predictive power, ROC-AUC was the primary evaluation metric. ROC-AUC was great in this situation because of its ability to differentiate days with positive returns over days with negative returns. Log-Loss was also important because of its ability to penalize confident but incorrect predictions.
- **Future Implications:**
  - Can calendar effects influence the magnitude of stock returns rather than their direction, and does this effect vary across companies?
  - How can the combination of calendar events along with macroeconomic announcements impact future stock performance?
  - Is there a relationship between calendar variables and changes in the distributional shape of returns, and do these variables exert greater influence on return behavior than on direction?

Input and Target Variables

‘return’	**The daily adjusted return of the stock.**
‘weekday’	**The day of the week. This is encoded.**
‘gap’	**Represents the overnight price change from the previous adjusted close price to the current open price.**
‘ret_5’	**Signifies the 5-day rolling average of returns. Used for identifying short-term momentum and reversion effects.**
‘vol_5’	**Represents the 5-day rolling average of standard deviations of returns. Represents short-term realized volatility.**
‘range_pct’	**The price dispersion relative to opening price that given day. Serves as a guiding principal for same-day volatility.**
‘volume_z’	**Standardization of trading volume relative to current history.**
'vol_change’	**Day-over-day changes in trading volume. Represents shift in market participation.**
‘Close’	**The closing price of the stock given that day.**
‘Target’ (Target Variable)	**Binary indicator equal to 1 or 0. A ‘1’ signifies the stock’s adjusted price increased when compared to the previous day. A ‘0’ represents that the stock’s adjusted price decreased compared to the previous day.**

Evaluation Metrics Used

Accuracy	**Fraction of days the model predicts the correct stock direction.**
ROC-AUC	**Measures whether the model ranks positive days above negative days, via stock direction. Represents the ability to distinguish between classes.**
Log-Loss	**Evaluates probability quality and penalizes overconfident wrong predictions.**

Can Calendar Signals Help Predict Next-Day Stock Direction for Apple? – Conclusion

Summary: This project revealed that there was no evidence supporting calendar effects as a reliable predictor for next-day stock direction in terms of price, volatility, and volume. Model performance was indistinguishable across all models and evaluation metrics. This was proven through accuracy, roc-auc, and log-loss being no better and even sometimes worse than random baselines. Daily stock direction appears to be dominated by noise in the short-term and calendar effects are too weak and unstable to be used as a real predictor of market prices. Overall, the results reveal the importance of using statistical methodology and domain expertise when trying to predict stock performance.

Key Takeaways: Across the Logistic Regression, Random Forest, and Gradient Boosting Models, performance was consistent. Accuracy and Roc-Auc both fell around the 50% mark, while Log-Loss was no better than the random benchmark of *0.69-0.70*. This reveals the model's inability to differentiate positive stock performance days versus negative stock performance days and produce informative probability estimates. This aligns with prior research indicating that calendar effects have minimal influence on short-term stock performance.

Next Steps: In the future, I think evaluating more of an index-level stock, such as the S&P 500, would give a more diversified investigation into this project. Moreover, it would be extremely interesting to explore tail risk, rolling-windows evaluations, return magnitude, or even realized volatility instead of binary price direction. Evaluating these targets might give us a better perspective into seeing the impact of calendar effects across different investment horizons and market regimes. I am looking forward to further investigating the impact of calendar signals with stock performance.